Large Sample Tests

The Central Limit Theorem and Z tests for proportions, means and standard deviations.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Central Limit Theorem

As n increases

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

- Large sample: generally $n \geq 30$.
- Needs **independent** observations and **finite variance**.
- **Standard error** = SD of a sampling distribution; for $\bar{x}$ it is $\sigma/\sqrt{n}$, which shrinks as n grows.

2 Z test decision rules

Test	Reject H_0 if	5%	1%
Two-tailed	$\lvert Z\rvert > Z_{\alpha/2}$	1.960	2.576
Right-tailed	$Z > Z_{\alpha}$	1.645	2.326
Left-tailed	$Z < -Z_{\alpha}$	−1.645	−2.326

State H_0 about the **population parameter**; compute Z from **sample statistics**.

3 Proportions

Single proportion, $H_0 : P = P_0$

$$Z = \frac{p - P_0}{\sqrt{P_0 Q_0 / n}}, \quad Q_0 = 1 - P_0$$

Two proportions, $H_0 : P_1 = P_2$

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Pooled $\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$, $\hat{Q} = 1 - \hat{P}$. Needs enough expected successes **and failures**.

4 Single mean, $H_0 : \mu = \mu_0$

σ known

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

σ unknown

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

With a large sample, s is a good substitute for σ.

5 Two means, $H_0 : \mu_1 = \mu_2$

Equal SDs

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Pooled σ if unknown

$$\sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}}$$

Unequal SDs

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

For independent samples, the variance of a difference is the **sum** of the variances.

6 Two SDs

Approximate test

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}$$

Under normal-population assumptions.

7 Choosing the Z test

Question	H_0	Standard error uses
Is the germination rate 80%?	$P = P_0$	P_0 (hypothesised)
Do two districts have the same adoption rate?	$P_1 = P_2$	Pooled $\hat{P}$
Is mean yield 40 q/ha?	$\mu = \mu_0$	σ or s
Do two varieties yield the same?	$\mu_1 = \mu_2$	Common or separate variances

! Common mistakes

- × Using p instead of P_0 in the single-proportion SE.
- × Ignoring the sign of Z in one-tailed tests.
- × Using 1.96 for a one-tailed 5% test (it's 1.645).

? Fill in the blanks

1. With σ = 10 and n = 25, the standard error of the mean is ———. [2]
2. x̄ = 52, μ_0 = 50 and SE = 1 give Z = ———, so a two-tailed test at 5% ——— H_0. [2; rejects]
3. With P_0 = 0.5, n = 100 and p = 0.6, the Z statistic is ———. [2]
4. Making the sample four times larger makes the standard error ———. [half as large]
5. Z = 1.8 in a right-tailed test at 5% means H_0 is ———. [rejected (1.8 > 1.645)]
6. Samples of 100 with p_1 = 0.4 and p_2 = 0.6 give a pooled proportion P̂ = ———. [0.5]

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